

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML – C1
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Based Pest Risk Prediction and Crop Protection Advisory System

Abstract

Problem Statement

Indian farmers face significant crop losses due to pest outbreaks that can spread rapidly before timely preventive action is taken. Pest occurrence is influenced by factors such as crop type, weather conditions, historical pest patterns, soil characteristics, and geographical location. Farmers, particularly those who lack timely access to agricultural information, may identify pest problems only after visible crop damage has occurred. Therefore, there is a need for an intelligent system that can assess pest risk in advance and support timely decision-making. The Ministry of Agriculture and Farmers Welfare has identified pest outbreak prediction, crop vulnerability identification, and timely crop-protection advisories as important AI applications for agriculture.

The proposed project aims to develop an **AI-based Pest Risk Monitoring and Crop Protection Advisory System** for farmers and agriculture field officers. The system will analyze multiple agricultural and environmental factors instead of depending on a single indicator. Pest data, rainfall and dry-spell information, Crop Registry, Digital Crop Survey, and Soil Health Card data identified will be considered for developing the solution. By combining these factors, the system will estimate the likelihood of pest outbreaks for specific crops and geographical regions.

The proposed system will provide **pest-risk prediction, crop vulnerability assessment, geographical risk visualization, early-warning alerts, and crop-specific protection advisories**. The AI model will classify areas or crops into appropriate risk levels and provide timely information to help farmers take preventive measures. Agriculture field officers can also use regional risk information to identify areas requiring attention. The project will focus on developing a predictive AI layer that converts agricultural data into actionable pest-risk information.

The scope is limited to pest-risk prediction, monitoring, alerts, and evidence-based advisories. It will not replace agricultural experts, independently prescribe pesticides, or guarantee that an outbreak will occur. The project's success will be evaluated using measurable AI/ML metrics such as **accuracy, precision, recall, and F1-score**, along with the usefulness and usability of the generated alerts and advisories.

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References (Category specific) :

1 . Kim, Y. H., Yoo, S. J., Gu, Y. H., Lim, J. H., Han, D., & Baik, S. W. (2014). Crop Pests Prediction Method Using Regression and Machine Learning Technology: Survey. IERI Procedia, 6, 52–56. <https://doi.org/10.1016/j.ieri.2014.03.009> (Base Paper)

2 . A. Tripathy et al., “Data Mining and Wireless Sensor Network for Agriculture Pest/Disease Predictions,” 2011 World Congress on Information and Communication Technologies (WICT), 2011. This work explores data mining and wireless sensor network approaches for predicting agricultural pest and disease occurrences, providing a foundation for data-driven pest monitoring and early prediction.
[https://IEEE/DOI – 10.1109/WICT.2011.6141424](https://IEEE/DOI-10.1109/WICT.2011.6141424)

3 . A. Amiri & A. R. Bandani, “Artificial Intelligence for Pest Identification and Decision Support in Sustainable Crop Protection: A Critical Review,” European Journal of Agronomy, 2026. This work is relevant to AI-based pest identification and crop-protection decision support, and discusses challenges associated with applying AI in sustainable crop protection, including dataset limitations and field-level generalization.
[https://IEEE/DOI – 10.1016/j.eja.2026.128040](https://IEEE/DOI-10.1016/j.eja.2026.128040)

4 . Amiri, A., & Bandani, A. R. (2026). Artificial Intelligence for Pest Identification and Decision Support in Sustainable Crop Protection: A Critical Review. European Journal of Agronomy.
<https://doi.org/10.1016/j.eja.2026.128040>

5 . A Model for Pest Infestation Prediction in Crops Based on Local Meteorological Monitoring Stations.” IEEE Conference Publication.
<https://ieeexplore.ieee.org/document/8575564>

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A61C5	S.V.D.DEEPTHI	
2	23BQ1A61D2	S.TEJASWI	
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Project Guide

Name:

Signature:

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.